

33-5 Reflection from metals

We can now use our results to understand the interesting phenomenon of reflection from metals. Why is it that metals are shiny? We saw in the last chapter that metals have an index of refraction which, for some frequencies, has a large imaginary part. Let's see what we would get for the reflected intensity when light shines from air (with $n = 1$) onto a material with $n = -in_f$. Then Eq. (33.55) gives (for normal incidence)

$$\frac{E'_0}{E_0} = \frac{1 + in_f}{1 - in_f}$$

For the *intensity* of the reflected wave, we want the square of the absolute values of E'_0 and E_0 :

$$\frac{I_r}{I_i} = \frac{|E'_0|^2}{|E_0|^2} = \frac{|1 + in_f|^2}{|1 - in_f|^2},$$

or

$$\frac{I_r}{I_i} = \frac{1 + n_f^2}{1 + n_f^2} = 1. \quad (33.58)$$

For a material with an index which is a pure imaginary number, there is 100 percent reflection!

Metals do not reflect 100 percent, but many do reflect visible light very well. In other words, the imaginary part of their indexes is very large. But we have seen that a large imaginary part of the index means a strong absorption. So there is a general rule that if *any* material gets to be a *very* good absorber at any frequency, the waves are strongly reflected at the surface and very little gets inside to be absorbed. You can see this effect with strong dyes. Pure crystals of the strongest dyes have a "metallic" shine. Probably you have noticed that at the edge of a bottle of purple ink the dried dye will give a golden metallic reflection, or that dried red ink will sometimes give a greenish metallic reflection. Red ink absorbs out the greens of *transmitted* light, so if the ink is very concentrated, it will exhibit a strong surface *reflection* for the frequencies of green light.

You can easily show this effect by coating a glass plate with red ink and letting it dry. If you direct a beam of white light at the back of the plate, as shown in Fig. 33-8, there will be a transmitted beam of red light and a reflected beam of green light.

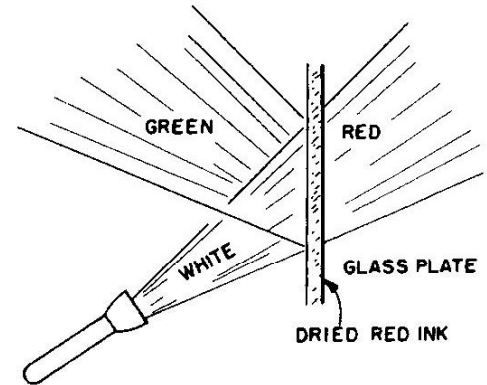


Fig. 33-8. A material which absorbs light strongly at the frequency ω also reflects light of that frequency.

da "The Feynman Lectures on Physics"
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